Exploiting the Structure of a Polynomial Optimization Problem

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Table of contents

Challenges

Geometric or standard form

Newton polytope

Conclusion

Challenges

Sum-of-Squares Programming

It may look simple at first...

$p(x)$ is SOS $\rightarrow p(x) = b(x)^\top Q b(x)$ where Q is PSD.

- PSD $\rightarrow$ (scaled) diagonally dominant ? *DSOS/SDSOS*
- **reformulation as geometric or standard conic form ?**
- **what polynomial space for $b(x)$? *Newton polytope***
- which basis for $b(x)$?
- which basis for $p - b^\top Q b$? Ill-conditioned change of basis ?
- any group symmetry ? Can we reduce symbolically ?
- *Chordal* sparsity ? *Term* sparsity ? *Sign* symmetry ?
- extract roots of $p(x)$ from dual moment matrix ?
- Different formulation ? Hypatia/Alphone, Burer-Monteiro ?

With constraints

$$p(x) = s_0(x) + \sum \lambda_i(x) h_i(x) + \sum s_i(x) g_i(x), \quad s_i(x) = b_i(x)^\top Q_i b_i(x)$$

- Explicit λ_i or remainder with Gröbner basis ?
- Putinar or Schüdgen certificate ?
- **what polynomial space for $b_i(x)$? *Newton polytope***
- which basis for $b_i(x)$?
- ...

Geometric or standard form

Geometric and standard form

Standard conic form SDP:

$$\begin{aligned} \min_{Q \in \mathcal{S}^n} \quad & \langle C, Q \rangle \\ \text{subject to:} \quad & \langle A_i, Q \rangle = b_i, \quad i = 1, 2, \ldots, m \\ & Q \succeq 0, \end{aligned}$$

Geometric conic form SDP:

$$\begin{aligned} \max_{y \in \mathbb{R}^m} \quad & \langle b, y \rangle \\ \text{subject to:} \quad & C \succeq \textstyle\sum_{i=1}^m A_i y_i \\ & y \text{ free}, \end{aligned}$$

Standard conic form

Notation

$$p(x) = \sum_{\alpha} p_\alpha x^\alpha \qquad \mathcal{A}_\alpha = \{(\beta, \gamma) \in b^2 \mid x^\beta x^\gamma = x^\alpha\}$$

Standard conic form

$$\langle \sum_{(\beta,\gamma)\in\mathcal{A}_\alpha} e_\beta e_\gamma^\top, Q \rangle = p_\alpha, \quad \forall \alpha$$

$$Q \succeq 0$$

Geometric conic form

Notation

$$p(x) = \sum_{\alpha} p_\alpha x^\alpha \qquad \mathcal{A}_\alpha = \{(\beta, \gamma) \in b^2 \mid x^\beta x^\gamma = x^\alpha\}$$

Let $(\beta_\alpha, \gamma_\alpha) \in \mathcal{A}_\alpha$.

Geometric conic form

$$\sum_{\alpha} p_\alpha e_{\beta_\alpha} e_{\gamma_\alpha}^\top + \sum_{(\beta,\gamma)\in\mathcal{A}_\alpha\setminus(\beta_\alpha,\gamma_\alpha)} y_{\beta,\gamma}(e_\beta e_\gamma - e_{\beta_\alpha} e_{\gamma_\alpha}^\top)^\top \succeq 0$$

$y_{\beta,\gamma}$ free

Which one do I choose ?

- Standard conic form is good when low number of variables and high degree. Univariate $2d$: standard gives linear $m = 2d + 1$ and geometric gives quadratic $m = d(d+1)/2 - (2d+1)$.
- Geometric conic form is good when high number of variables and low degree. Quadratic: standard gives quadratic m and geometric gives $m = 0$.

What's the threshold used in practice ? None, user chooses !

Which one do I choose ?

- Standard conic form is good when low number of variables and high degree. Univariate $2d$: standard gives linear $m = 2d + 1$ and geometric gives quadratic $m = d(d+1)/2 - (2d+1)$.
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What's the threshold used in practice ? None, user chooses !

In YALMIP, `sosmodel` uses geometric and `solvesos` uses standard.

In `SumOfSquares.jl`, formulation matches solver's conic form !
Hence the importance of playing with `Dualization.jl` !

Newton polytope

Newton polytope

Notation

$$p(x) = \sum_{\alpha} p_\alpha x^\alpha \qquad \mathcal{A}_\alpha = \{(\beta, \gamma) \mid x^\beta x^\gamma = x^\alpha\}$$

Key observation 1

If $\mathcal{A}_\alpha = \{(\beta, \beta)\}$ and $p_\alpha = 0$ then $Q_{\beta,\beta} = 0$.

Key observation 2

If $Q_{\beta,\beta} = 0$ then $Q_{\beta,\gamma} = Q_{\gamma,\beta} = 0$, $\forall \gamma$.

Newton chip method

while $\exists \alpha, \beta$ s.t. $\mathcal{A}_\alpha = \{(\beta, \beta)\}, p_\alpha = 0$ **do**
  remove β from b
end while

- Easy to implement
- Easy to generalize, e.g., [BKP16, Section 2.3][1]
- Costly : quadratic in length of $b \longrightarrow$ best trim down b before
- Signed variant: change $p_\alpha = 0$ into $p_\alpha \leq 0$

[BKP16] Burgdorf, Klep, and Povh. *Optimization of polynomials in non-commuting variables*. 2016.

Newton polytope

Polytopic reformulation of key observation 2

$\mathcal{N}(b) + \mathcal{N}(b) = 2\mathcal{N}(b)$ where $+$ is Minkwoski sum.

So the result of the commucative unsigned newton chip method is:

$$\mathcal{N}(b) = \lfloor \mathcal{N}(p)/2 \rfloor$$

- Less costly because not quadratic in length of b
- How to enumerate the elements given V-rep of $\lfloor \mathcal{N}(p)/2 \rfloor$?
  - solve LP's [Lof09][2]
  - Compute H-rep with CDD/LRS/PPL [Pra+04][3]

[Lof09] Lofberg. "Pre-and post-processing sum-of-squares programs in practice". 2009.

[Pra+04] Prajna et al. "New developments in sum of squares optimization and SOSTOOLS". 2004.

Wait, isn't polyhedral computation not cheap ?

If polyhedral computation is costly, we might as well just do Newton chip method!

We can at least compute a cheap outer approximation of

$$\lfloor \mathcal{N}(p)/2 \rfloor$$

Outer-approximation of [Pra+04][4]

- Min and max total degree
- Min and max degree of each variable
- Min and max degree groups (*multipartite*)

[Pra+04] Prajna et al. "New developments in sum of squares optimization and SOSTOOLS". 2004.

Support function interpretation

$2\mathcal{N}(b) \subseteq \mathcal{N}(p)$ is equivalent to

$$\forall y \in \mathbb{R}^n, 2\delta^*(y|\mathcal{N}(b)) \leq \delta^*(y|\mathcal{N}(p)).$$

Generalizes previous cases:

- Max total degree: $y = \mathbf{1}$
- Min total degree: $y = -\mathbf{1}$
- Max degree of x_i: $y = e_i$
- Min degree of x_i: $y = -e_i$
- Max degree group $\{x_{i_1}, \ldots, x_{i_j}\}$, $y = e_{i_1} + \cdots + e_{i_j}$
- Min degree group $\{x_{i_1}, \ldots, x_{i_j}\}$, $y = -e_{i_1} - \cdots - e_{i_j}$

Signed Newton chip for Putinar

Notation

$$g_0 = 1 \qquad \mathcal{A}^i_\alpha = \{(\beta,\gamma) \in b_i^2 \mid x^\beta x^\gamma = x^\alpha\}$$

Generalized key observation 1

If

$$\nexists i, \beta \neq \gamma \in b_i, \delta \in \mathcal{N}(g_i) \text{ s.t. } x^\delta x^\gamma x^\beta = x^\alpha$$

and

$$\{g_{i,\delta} \mid \exists \beta_i \in b_i \text{ s.t. } x^\delta x^{2\beta_i} = x^\alpha\}$$

has constant sign and p_α is zero or has opposite sign, then $Q_{i,\beta_i,\beta_i} = 0$

Support function for Putinar

$$\forall y \in \mathbb{R}^n, \delta^*(y|\mathcal{N}(p)) = \delta^*(y|\mathcal{N}(\sum g_i s_i)).$$

Could be cancellations if negative coefficients in g_i

$$\forall y \in \mathbb{R}^n, \delta^*(y|\mathcal{N}(p)) \subseteq \delta^*(y|\sum \mathcal{N}(g_i) + \mathcal{N}(s_i)).$$

Support function on Minkowski sum:

$$\delta^*(y|\mathcal{S}+\mathcal{T}) = \delta^*(y|\mathcal{S}) + \delta^*(y|\mathcal{T})$$

$$\begin{aligned}\forall y \in \mathbb{R}^n, \delta^*(y|\mathcal{N}(p)) &\subseteq \sum \delta^*(y|\mathcal{N}(g_i)) + \delta^*(y|\mathcal{N}(s_i)) \\ &\subseteq \sum \delta^*(y|\mathcal{N}(g_i)) + 2\delta^*(y|\mathcal{N}(b_i))\end{aligned}$$

If support function uniquely maximized by diagonal elements $Q_{i_1}, \ldots, Q_{i_j}$ with same signs $g_{i_1}, \ldots, g_{i_j}$. If $\delta^*(y|\mathcal{N}(p))$ strictly smaller or different sign, reduce by removing $b_{i_1}, \ldots, b_{i_j}$.

Conclusion

Newton polytope vs sparsity/symmetry

- Newton polytope reduces the basis b
- Sparsity/symmetry try to block diagonalize b, but b is not reduced,

Complementary reductions, a good Newton polytope reduction is as important as sparsity/symmetry reduction.

Conclusion

- Always try to solve your problem with and without `Dualization.jl`
- Specify constraints with `domain` keyword to get Newton polytope-reduced bases.
- Implemented in `SumOfSquares.jl`, feedback is welcome!

Algebraic geometry

- Interface `MultivariatePolynomials.jl` with implementations
  - `DynamicPolynomials.jl`
  - `TypedPolynomials.jl`
  - `SIMDPolynomials.jl`
  - `CondensedMatterSOS.jl`
- Polynomials bases in `MultivariateBases.jl`
- Gröbner bases and algebraic system solving in `SemialgebraicSets.jl`. Interface with Buchberger and multiplication matrices by default with other implementations:
  - `HomotopyContinuation.jl`
  - `Groebner.jl`
- Extract roots from moment matrix with `MultivariateMoments.jl`

Optimization with JuMP

- Solver interface `MathOptInterface.jl` implemented by (SDP-only):
  - Nonsymmetric cone interior point: `Hypatia.jl`
  - `MosekTools.jl`
  - ADMM: `SCS.jl`, `COSMO.jl`
  - MATLAB: `SeDuMi.jl`, `SDPNAL.jl`, `SDPT3.jl`
  - C/C++: `CSDP.jl`, `SDPA.jl`, `DSDP.jl`
  - Burer-Monteiro: `SDPLR.jl`
  - BMI, NLSDP: `Penopt.jl`
- `JuMP.jl` extension for optimization with polynomials `PolyJuMP.jl`
  - Solve KKT system with `SemialgebraicSets.jl`
  - Sums of AM/GM Exponential (SAGE) : `https://github.com/jump-dev/SumOfSquares.jl/pull/240/`
  - Sum-of-Squares with `SumOfSquares.jl`

